

Seshank Irukulapati

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EDUCATION

- PES University (PESIT)** Bengaluru, India
Bachelor of Technology - Computer Science; CGPA: 8.15 *August 2019 - July 2023*

SUMMARY

Results-driven Software Engineer with hands-on experience in building and optimizing scalable backend systems, distributed data pipelines, and high-performance microservices. Proficient in **Java, Spring Boot, Kafka, Spark and Big Data** technologies (Trino, Hive, S3). Adept at solving complex engineering challenges, improving system reliability, and driving performance optimizations across large-scale enterprise platforms.

SKILLS SUMMARY

- Languages:** Java, JavaScript, Python, SQL, Bash
- Frameworks:** Spring Boot, Hibernate, Angular
- Databases & Caching:** PostgreSQL, Apache Ignite, Elasticsearch, Redis, MongoDB, MySQL, MSSQL
- DevOps & Tools:** Docker, Kubernetes, Git, Grafana, Gradle, SonarQube, Jenkins
- Big Data & Cloud:** Apache Kafka, Spark, Hive, Trino, AWS S3, MinIO, Azure Blob Storage, GCS Blob Storage

EXPERIENCE

- Subex Limited** *October 2024 - present*
Senior Software Engineer
 - Engineered a high-performance match system in **Java** using Tries. Developed a custom **cache** implementation with scheduled updates and used Spark UDF for distributed processing with throughput of **4000+ records/second/core**.
 - Revamped case processing module using **Kafka** consumer pipeline with parallel processing, transactional retries, offset tracking in PostgreSQL database. Certified to process **1.2 million** cases per day.
 - Orchestrated seamless migration of **30TB** of customer data using Saga pattern across multiple phases with limited downtime, implementing fault tolerance and rollback mechanisms.
 - Optimized **Spark** streaming pipelines through salting, dynamic allocation, and phased aggregations — **reducing runtime and memory usage by 20%**.
 - Led production stabilization efforts, **resolved 150+** critical issues and delivered key optimizations improving performance and system reliability, while maintaining SLA compliance.
 - Designed and integrated **MySQL** and **MSSQL** support alongside PostgreSQL enabling direct integration with customer databases containing millions of rows of live data and eliminating daily synchronization overhead.
- Subex Limited** *August 2023 - October 2024*
Software Engineer
 - Implemented multi-cloud partitioning and backup strategy across S3, Azure Blob, GCS Blobs, PostgreSQL and Elasticsearch. Created data cleanup policy usable on all tables across the platform **Reducing storage costs by 30%**.
 - Improved the performance of telemetry system by **60%**, reducing data gathering time from **4 minutes to 50 seconds** by parallelizing API calls, refactoring code and optimizing SQL queries in dependent microservices.
 - Developed a DAG Rebuilder to run pre-processing logic, reconstruct, and optimize pipeline execution graphs, eliminating redundant node processing, **improving performance by 40%**.
 - Designed and implemented a **Kafka**-based task execution framework enabling real-time tracking across all application modules. Built a 30-day audit ledger to re-trigger tasks and maintain audit trails.
 - Implemented a configurable Spark-based summarization framework supporting dynamic filters, group-by dimensions, aggregation functions, and time-window logic for **Fraud Management**.
 - Developed and optimized RESTful Web Services in a microservice architecture using Java, Spring Boot and Hibernate, ensuring efficient data access and management. Improved REST API response by **30%** by introducing cache requests.

PROJECTS

- Twitter Trends Analysis:** Developed a real-time Twitter stream processing pipeline, Integrated Twitter API to stream live tweets using Spark-Structured streaming and sent the tweets to Kafka and used MongoDB to consume, store and categorize tweets, enabling efficient aggregation and analysis. Tech: Python, Spark, Kafka, MongoDB

HONORS AND AWARDS

- Collaborative Concorde Award: Team award for outstanding contribution to product engineering and innovation, 2024
- Placed 6th for our project on analyzing green cover in cities at CMRIT Hackathon, 2019
- Honoured with Spot Award (twice) for my contributions to the product releases, 2024